

D672

Elementary Science and Engineering Methods

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A fourth-grade teacher begins a unit on erosion. The school just repaved its parking lot, and the unit runs for three weeks. Which launch best reflects three-dimensional, phenomenon-driven learning?

- A. Teach one science practice, graphing, in isolation.
- B. Start with a vocabulary list and textbook reading, then show the photos at the end.
- C. Show photos of a local hillside that washed away after a storm and ask students what might explain it, then plan investigations.
- D. Have students measure the parking lot.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

Students must design a bridge that holds at least 500 g, using only 20 craft sticks and 1 meter of tape, in 40 minutes. The class is named after a famous engineer, and the room is carpeted. Which is a constraint?

- A. Using only 20 craft sticks.
- B. Every requirement is a constraint.
- C. The carpeted room.
- D. Holding at least 500 g.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A design challenge lists: must keep an ice cube frozen for 30 minutes; cost under \$2; fit in a cup. Which item is a criterion?

- A. Cost under \$2.
- B. Fit in a cup.
- C. Build it in teams.
- D. Keep an ice cube frozen for 30 minutes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A group's paper-towel water filter lets muddy water through. The group named their filter "Clear Stream" and decorated it with stickers. What is the best next step?

- A. Analyze where mud passed through, change one layer, and test again.
- B. Start a new topic, since the design failed.
- C. Change all the materials at once.
- D. Add more stickers to improve it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Students test whether ramp height affects how far a car rolls. Group A uses a 20 cm ramp and a heavy car; group B uses a 40 cm ramp and a light car. The ramps are painted blue and the tests happen in the gym. What is the problem?

- A. Distance should be the independent variable.
- B. Both ramp height and car mass changed, so the effect of height cannot be isolated.
- C. The ramps should be different colors.
- D. Nothing; more differences give more data.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

In a plant experiment, students vary the amount of light and measure plant height. Which variable should they keep the same?

- A. None; only light matters.
- B. Plant height.
- C. The amount of light.
- D. The amount of water each plant receives.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A student says, "Evolution is just a theory, so it has not been proven like a law." The student's older sister studies biology at a university. Which response best reflects the nature of science?

- A. A scientific theory is a well-tested explanation; theories do not turn into laws, which describe patterns.
- B. Scientific knowledge never changes.
- C. Right, theories become laws once proven.
- D. Ask the sister, since she studies biology.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which statement describes a scientific law rather than a theory?

- A. Matter is made of tiny particles, explaining its properties.
- B. Living things share common ancestors, explaining their similarities.
- C. Objects attract each other with a force that depends on their masses and distance.
- D. A hypothesis that has been proven.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A student writes: "Dark cups got hotter. The black cup reached 38°C and the white cup 29°C." She used neat handwriting and a ruler for her table. What does her explanation need most?

- A. Neater handwriting.
- B. More data.
- C. Reasoning that explains, using a science idea, why dark colors absorb more light energy.
- D. Restating the temperatures again.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A student's reasoning reads: "Plants need light to make food through photosynthesis, so less light means less growth." What claim does this reasoning best support?

- A. Light has no effect on growth.
- B. Plants grown with more light grew taller.
- C. Plant A was 12 cm and plant B was 7 cm.
- D. Plants grow better with more water.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

During a discussion about why puddles disappear, one student says, “The sun drinks the water.” The class has 24 students and the discussion is before lunch. Which teacher response best supports productive discourse?

- A. “No, that's wrong. It's evaporation.”
- B. “We'll talk after lunch.”
- C. “Good job!” and move on.
- D. “Can you say more about what you mean? Who can add to or question that idea?”

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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12.

A second-grade class will launch small rubber-band rockets outside. The activity takes only 10 minutes, and the students did something similar last year. What should the teacher do first?

- A. Skip rules since students did it before.
- B. Require safety goggles, set a launch zone, and review rules before any launch.
- C. Skip goggles because it is short.
- D. Let students choose whether to wear goggles.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A fourth-grade teacher begins a unit on erosion. ~~The school just repaved its parking lot, and the unit runs for three weeks.~~ Which launch best reflects three-dimensional, phenomenon-driven learning?

Target word: phenomenon-driven launch **Decides:** Start with a puzzling real event students work to explain.

A	Dropped element	Drops the phenomenon and the other dimensions.
B	Inverted operation	Puts facts before sense-making.
C	Correct	An anchor phenomenon drives use of practices and ideas.
D	Camouflage	Reacts to the repaving; not an erosion phenomenon.

2. Answer A Concept 2

Students must design a bridge that holds at least 500 g, using only 20 craft sticks and 1 meter of tape, in 40 minutes. ~~The class is named after a famous engineer, and the room is carpeted.~~ Which is a constraint?

Target word: criteria vs. constraints **Decides:** Constraints are limits.

A	Correct	A limit on materials is a constraint.
B	Absolute language	Criteria and constraints are different.
C	Camouflage	Not part of the design problem.
D	Confusion set	That is a criterion, what success means.

3. Answer D Concept 2 · transfer

A design challenge lists: must keep an ice cube frozen for 30 minutes; cost under \$2; fit in a cup. Which item is a criterion?

Target word: reverse: find the criterion **Decides:** Criteria describe success.

A	Inverted operation	A budget limit is a constraint.
B	Confusion set	A size limit is a constraint.
C	Adjacent category	A classroom procedure, not a design requirement.
D	Correct	Describes what the design must accomplish.

4. Answer A Concept 3

A group's paper-towel water filter lets muddy water through. ~~The group named their filter "Clear Stream" and decorated it with stickers.~~ What is the best next step?

Target word: iteration **Decides:** Analyze the failure and make one change.

A	Correct	Iterative redesign uses failure as data.
B	Dropped element	Drops the redesign cycle.
C	Adjacent category	Too many changes hide what helped.
D	Camouflage	Decoration does not affect function.

5. Answer B Concept 4

Students test whether ramp height affects how far a car rolls. Group A uses a 20 cm ramp and a heavy car; group B uses a 40 cm ramp and a light car. ~~The ramps are painted blue and the tests happen in the gym.~~ What is the problem?

Target word: fair test **Decides:** Two variables changed.

A	Inverted operation	Distance is the measured, dependent variable.
B	Correct	Only one independent variable should change.
C	Camouflage	Color is not the issue.
D	Absolute language	Uncontrolled variables confuse results.

6. Answer D Concept 4 • transfer

In a plant experiment, students vary the amount of light and measure plant height. Which variable should they keep the same?

Target word: reverse: find a controlled variable **Decides:** Light is independent; height is dependent.

A	Dropped element	Drops the need for controls.
B	Confusion set	That is the dependent variable.
C	Inverted operation	That is the independent variable.
D	Correct	Must be controlled so only light differs.

7. Answer A Concept 5

A student says, "Evolution is just a theory, so it has not been proven like a law." ~~The student's older sister studies biology at a university.~~ Which response best reflects the nature of science?

Target word: theory vs. law **Decides:** Theories explain; laws describe; neither becomes the other.

A	Correct	Theory and law are different kinds of knowledge.
B	Absolute language	It is durable but open to revision.
C	Confusion set	A common misconception.
D	Camouflage	Authority is not the point.

8. Answer C Concept 5 • transfer

Which statement describes a scientific law rather than a theory?

Target word: reverse: identify a law **Decides:** Laws describe patterns.

A	Adjacent category	Particle theory is an explanation.
B	Inverted operation	An explanation, a theory.
C	Correct	Describes a regular pattern (law of gravitation).
D	Confusion set	Laws are not graduated hypotheses.

9. Answer C Concept 6

A student writes: "Dark cups got hotter. The black cup reached 38°C and the white cup 29°C." ~~She used neat handwriting and a ruler for her table.~~ What does her explanation need most?

Target word: missing CER part **Decides:** She has a claim and evidence but no science reasoning.

A	Camouflage	Presentation does not complete the explanation.
B	Adjacent category	She already has evidence.
C	Correct	Reasoning connects evidence to the claim.
D	Dropped element	Restated data is not reasoning.

10. Answer B Concept 6 • transfer

A student's reasoning reads: "Plants need light to make food through photosynthesis, so less light means less growth." What claim does this reasoning best support?

Target word: reverse: from reasoning to claim **Decides:** The reasoning links light to growth.

A	Inverted operation	Opposite of the reasoning.
B	Correct	The reasoning explains a light-growth claim.
C	Confusion set	That is evidence, not a claim.
D	Adjacent category	Different variable.

11. Answer D Concept 7

During a discussion about why puddles disappear, one student says, "The sun drinks the water." ~~The class has 24 students and the discussion is before lunch.~~ Which teacher response best supports productive discourse?

Target word: talk move **Decides:** Push students to reason and respond to each other.

A	Inverted operation	Teacher evaluation ends student thinking.
B	Camouflage	Timing is not the issue.
C	Dropped element	Praise without probing drops the reasoning.
D	Correct	Presses for reasoning and invites peer response.

12. Answer B Concept 8

A second-grade class will launch small rubber-band rockets outside. ~~The activity takes only 10 minutes, and the students did something similar last year.~~ What should the teacher do first?

Target word: safety priority **Decides:** Flying objects require eye protection and rules.

A	Absolute language	Experience does not remove the need for safety.
B	Correct	Safety is set before hands-on work begins.
C	Camouflage	Duration does not reduce the risk.
D	Adjacent category	Safety equipment is required, not optional.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	8	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Three-dimensional, phenomenon-driven learning	Students make sense of a real phenomenon by using science and engineering practices, crosscutting concepts, and core ideas together, rather than learning facts first.	Teaching vocabulary and facts first and saving the phenomenon for the end, or treating a single practice as the whole lesson.	How a unit is launched and structured.
2	Engineering criteria and constraints	Criteria are what a successful design must do; constraints are limits such as materials, cost, and time.	Swapping the two terms.	Requirements and limits listed for a design challenge.
3	Iterative design	Engineers test, analyze why a design failed, change one thing, and retest; failure is useful data.	Treating a failed prototype as the end of the task or changing many things at once.	A prototype that did not meet the criteria.
4	Fair tests and variables	Change one independent variable, measure the dependent variable, and keep other variables controlled.	Changing several variables at once.	An investigation plan with more than one difference between trials.
5	Nature of science: theories and laws	A law describes a pattern (often mathematically); a theory is a well-tested explanation of why. Theories do not become laws, and scientific knowledge can be revised with new evidence.	Thinking a theory is a guess that becomes a law once proven.	Use of the words theory, law, or proven.
6	Claim, evidence, reasoning	A scientific explanation makes a claim, backs it with data, and uses a science idea to explain why the evidence supports it.	Treating restated data as reasoning or skipping the science idea.	Assessing a student's written explanation.
7	Productive science discourse	Talk moves such as asking students to explain their reasoning, add on, or respond to a peer keep students doing the thinking.	The teacher evaluating each answer right away as right or wrong.	A class discussion of competing ideas.
8	Safety first in hands-on science	Safety procedures (goggles, supervision, safe materials, clear rules) are set and followed before any hands-on activity starts.	Skipping safety steps for short or familiar activities.	Heat, chemicals, sharp tools, or flying objects in a lesson.

Course scope (public description): Elementary Science and Engineering Methods is a dynamic and forward-thinking course designed for educators who aim to integrate science and engineering practices into their elementary classroom. This course focuses on the foundational methods of teaching science and engineering concepts, emphasizing hands-on, experiential learning. Students explore effective strategies for fostering inquiry, creativity, and critical thinking among young learners through science experiences, engineering projects, and the use of technology. The curriculum covers key topics such as the three-dimensional learning approach and scientific literacy and the nature of science. Through intentional instructional decision-making and reflective practice, educators learn to create an inclusive, stimulating, and safe learning environment that encourages students to explore, question, innovate, and participate in productive discourse in the fields of science and engineering.